

PulseRoute: An Inclusive Contactless Vital Signs Monitoring System with Adaptive Face-to-Palm Signal Routing for Teleconsultation

Arij Belmabrouk

Independent Researcher

arij.belmabrouk@example.com

Abstract

Contactless photoplethysmography (rPPG) enables camera-based measurement of heart rate (HR), heart rate variability (HRV), and respiratory rate (RR) without physical contact. However, existing systems are predominantly validated on light-skinned subjects under controlled laboratory conditions, resulting in systematic measurement failures for individuals with darker skin tones, in low-ambient lighting, or in real-world clinical environments. We present PulseRoute, a contactless vital signs monitoring system designed for teleconsultation settings that addresses these limitations through two novel contributions. First, a per-session subject calibration mechanism that replaces all fixed population-average thresholds with dynamically computed personal thresholds derived from the patient's own signal during a 10-second setup phase. Second, an adaptive signal routing mechanism that begins measurement from the face (for better user experience) and automatically routes to the palm when the face signal is insufficient — a decision informed by both a composite signal quality score and a physics-based amplitude floor derived from our pilot data. Pilot evaluation on two subjects across Fitzpatrick skin types IV–VI demonstrates correct HR estimation (± 0.5 – 2.5 BPM against pulse oximeter reference), accurate respiratory rate detection (14–20 BrPM), and correct palm routing triggered for FST VI subjects under standard home lighting conditions where existing face-only systems would produce silently inflated HRV measurements. PulseRoute is implemented in Python using BiSeNet face parsing, the POS algorithm, and an 11-step modular pipeline, and is made publicly available.

Keywords: remote photoplethysmography, contactless vital signs, heart rate variability, skin tone bias, teleconsultation, signal routing, facial ROI, palm ROI

1. Introduction

The COVID-19 pandemic accelerated the adoption of telemedicine and remote patient monitoring, creating urgent demand for contactless physiological measurement tools that can operate in uncontrolled home environments. Remote photoplethysmography (rPPG) — the extraction of blood volume pulse signals from video of the skin surface — offers a compelling solution: it requires only a standard webcam, no wearable hardware, and no physical contact [1].

Existing rPPG systems have demonstrated reliable heart rate estimation under laboratory conditions, with several commercial implementations achieving mean absolute errors below 5 BPM [2, 3]. However, these systems share a critical limitation: their design assumptions, threshold values, and validation datasets are predominantly drawn from studies on light-skinned subjects under controlled lighting [4, 5]. This creates a documented performance gap for individuals with darker skin tones — Fitzpatrick skin types (FST) IV through VI — who represent a substantial fraction of the global population and are often disproportionately represented among patients with conditions requiring remote monitoring (hypertension, cardiovascular disease, diabetes) [6].

The physical basis of this gap is well understood. Melanin in the epidermis absorbs a portion of incident light before it reaches the capillaries, reducing the amplitude of the reflected pulse signal [7]. At FST V–VI under typical home webcam conditions, the filtered pulse signal standard deviation can be 5–15 times weaker than the equivalent signal from FST I–II subjects. Fixed amplitude thresholds calibrated for laboratory subjects will classify these signals as poor quality and either discard the measurement or report inflated HRV metrics — without informing the clinician that the measurement is unreliable.

A second underexplored direction is the use of palm skin for rPPG signal extraction. Research has shown that the palm produces a 31% stronger rPPG signal than the face due to its higher density of superficial capillaries and significantly lower melanin content regardless of overall skin tone [8]. Despite this, no widely deployed rPPG system implements palm measurement as a clinical fallback, and the routing logic that would decide when to switch from face to palm has not been systematically studied.

We present PulseRoute, a system that addresses both problems. Our contributions are:

1. A **per-session subject calibration mechanism** (SubjectProfile) that replaces every fixed population-average threshold in the pipeline with a value derived from the patient's own signal during a 10-second setup phase, making the system adaptive across skin tones, lighting conditions, and camera qualities.
2. An **adaptive face-to-palm routing mechanism** that begins measurement from the face for better user experience and automatically routes to the palm when the face signal is insufficient, using a composite five-component quality score combined with a physics-based amplitude floor derived from empirical pilot data.
3. A **motion-robust recording protocol** that silently rejects artifact frames caused by patient coughing, talking, or repositioning, extending the recording window to accumulate the required number of clean frames rather than failing on a fixed time budget.

The system is designed specifically for teleconsultation settings where the patient is a real sick person sitting in their living room under ceiling lighting — not a healthy volunteer in a lab.

2. Related Work

2.1 Remote Photoplethysmography

The foundational rPPG work of Verkruijsse et al. [9] demonstrated that the green channel of standard RGB video contains a measurable blood volume pulse signal. Subsequent algorithmic advances include CHROM [10], which uses chrominance-based signal separation, and POS (Plane-Orthogonal-to-Skin) [11], which projects the normalized RGB signals onto the plane orthogonal to the skin color vector and achieves superior performance across different lighting conditions. PulseRoute uses the POS algorithm for its adaptive alpha weighting, which unlike CHROM requires no fixed skin color coefficients and therefore performs more equitably across Fitzpatrick types.

2.2 Facial Region of Interest Extraction

Early rPPG systems used simple rectangular face bounding boxes. More recent work uses facial landmark detection (MediaPipe [12]) or semantic segmentation (BiSeNet [13]) to extract specific skin regions. BiSeNet provides pixel-level classification of 19 facial regions, enabling precise exclusion of non-skin areas (hair, eyes, nose, lips) that degrade signal quality. PulseRoute uses BiSeNet V1 with a ResNet-18 backbone for forehead and cheek ROI extraction.

2.3 Skin Tone and rPPG Performance

Ba et al. [4] demonstrated significant performance degradation of rPPG systems at darker skin tones, with HR mean absolute error increasing from 2.1 BPM at FST I-II to 8.7 BPM at FST V-VI under equivalent conditions. Nowara et al. [5] showed that standard rPPG datasets are severely imbalanced toward lighter skin tones, with FST V-VI subjects comprising less than 8% of most benchmark datasets. The ITA (Individual Typology Angle) metric, derived from CIELab colorspace measurements of the ROI pixels, has been validated as a reliable proxy for Fitzpatrick skin type in rPPG contexts [14].

2.4 HRV from rPPG

Extraction of HRV metrics from rPPG requires accurate beat-to-beat interval (RR interval) estimation. McDuff et al. [15] showed that short-term RMSSD from rPPG achieves acceptable agreement with ECG gold standard when filtered signal amplitude is sufficient and the recording is at least 60 seconds. At lower signal amplitudes — precisely the condition encountered with darker skin — RR interval timing errors of ± 50 –100 ms have been reported, which cascade into RMSSD inflation of 150–300% above true values [16].

2.5 Palm-Based rPPG

Luo et al. [8] demonstrated that the palm produces a 31% stronger rPPG signal than the face, attributed to higher capillary density in the palmar dermis and the near-zero melanin content of the palm regardless of dorsal skin tone. Despite this finding, no clinical rPPG system implements palm measurement as a systematic fallback, and the decision boundary for when face measurement is insufficient has not been formally characterized.

3. System Design

3.1 Architecture Overview

PulseRoute implements an 11-step modular pipeline. The pipeline is described below and summarized in Figure 1.

Step 1 — Camera Initialization: The system initializes the webcam and measures actual fps over a 5-second warm-up window. Nominal fps values from camera drivers are rejected; only the empirically measured rate is used for all downstream temporal calculations. Minimum requirements: 15 fps, 480p resolution.

Step 2 — Face ROI Extraction and Subject Calibration (10 seconds): BiSeNet parses the face frame-by-frame to build a stable forehead and cheek mask. The first 5 seconds establish the mask; the subsequent 5 seconds sample pixel values through the locked mask to build a SubjectProfile (Section 3.2). The ITA angle is computed from the mean CIELab L^* and b^* values of the ROI pixels.

Step 3 — RGB Signal Extraction (35 seconds, adaptive): Per-frame mean R, G, B values are extracted from the combined forehead and cheek mask with ITA-adaptive pixel quality filtering. Frames where the green channel deviates by more than the patient's personal motion threshold from the rolling mean of the previous 10 frames are silently rejected as motion artifacts. The recording extends automatically to compensate for rejected frames, capped at $2.5\times$ the target duration.

Step 4 — Normalization: Each channel is divided by its temporal mean to remove DC offset (absolute brightness), then linearly detrended to remove slow lighting drift. Output is centered at zero.

Step 5 — POS Pulse Extraction: The POS algorithm [11] combines the three normalized channels into a single pulse waveform using an adaptive weight $\alpha = \sigma(S1)/\sigma(S2)$, where $S1 = R - G$ and $S2 = R + G - 2B$. The adaptive weighting requires no fixed skin color assumption.

Step 6 — Bandpass Filtering: A zero-phase Butterworth filter of order 4 is applied. The lower cutoff is set adaptively: if respiratory rate is known from Step 10 with confidence > 0.4 , the cutoff is placed 0.1 Hz above the breathing frequency; otherwise it defaults to 0.917 Hz (55 BPM). A pre-filter clipping step removes samples beyond $\pm 4\sigma$ to prevent filter ringing from artifact spikes.

Step 7 — FFT Frequency Analysis: The real FFT of the filtered signal produces a power spectrum from which the dominant HR frequency is identified.

Step 8 — Peak Detection: The dominant HR frequency from Step 7 gates an overlapping window search (80% advance) across the time-domain signal. Deduplication uses a local SNR + prominence score so that when two candidate beats are too close, the one with better local signal quality is kept rather than the earlier one. A gap-filling pass inserts missed beats in double-length intervals. An FFT cross-check overrides Step 8's frequency estimate if it deviates more than 15 BPM below the Step 7 dominant peak, which catches sub-harmonic selection errors.

Step 9 — HR and HRV Calculation: Final HR is a weighted combination (0.7 FFT, 0.3 time-domain) when the two estimates agree within 10 BPM. RMSSD and SDNN are computed from RR intervals. HRV interpretation uses age-adjusted norms when patient age is available.

Step 10 — Respiratory Rate: The normalized green channel is bandpass filtered to 0.1–0.5 Hz (6–30 BrPM) and the dominant frequency extracted via FFT with quadratic interpolation for sub-bin precision. This step runs before Step 5 to capture the breathing frequency before POS processing removes low-frequency content.

Step 11 — Signal Quality Score and Routing Decision: A composite score combining five metrics (spectral SNR, SNR in dB, RR regularity, personal amplitude score, HR confidence) produces a single quality score. Routing thresholds are adjusted downward by 5–15% for FST IV–VI subjects to prevent unfair routing caused by physics rather than measurement error. A physics-based amplitude floor (filtered std < 0.002) independently triggers palm routing regardless of the composite score, based on our empirical observation that beat timing errors at this amplitude level inflate RMSSD regardless of algorithm improvements.

3.2 SubjectProfile: Per-Session Dynamic Calibration

The SubjectProfile is the central architectural innovation of PulseRoute. It is computed once per session during Step 2 and passed through all downstream steps.

During the 5-second calibration phase, per-frame green channel means are collected through the locked ROI mask. The following personal thresholds are computed:

Motion threshold: $5 \times \text{baseline_g_std}$ (minimum 2.0). A frame whose green channel deviates from the rolling mean by more than this value is classified as a motion artifact. Using the patient's own noise floor means a naturally active patient has a proportionally higher threshold rather than being penalized by a threshold calibrated for still laboratory subjects.

Amplitude target: $0.004 \times 0.80 \times \text{baseline_g_std}$. This bridges the scale between raw green channel std (~0.3–2.0) and the filtered pulse signal std (~0.0005–0.005) through an empirically derived scaling factor. Step 11's amplitude score uses this as the ceiling so darker-skinned patients are scored against their own realistic best rather than a lab benchmark they cannot reach.

Bandpass hint: A rough HR estimate from the calibration signal is used to narrow the bandpass window to ± 30 BPM around the estimated rate, reducing noise from irrelevant frequency bands.

RR filter tolerance: Dynamically set between 0.30 (tight, clean signal) and 0.50 (loose, noisy signal) based on the composite quality score from Step 11.

3.3 Routing Logic

The routing decision in Step 11 operates through two independent checks:

Check 1 — Composite score: If the weighted five-component score falls below the ITA-adjusted threshold (default HIGH = 0.60, reduced to 0.50 for FST V and 0.45 for FST VI), the system routes to palm.

Check 2 — Amplitude floor: If the filtered signal standard deviation is below 0.002, the system routes to palm regardless of the composite score. This threshold is derived from our pilot data showing consistent RMSSD inflation above 200 ms at this amplitude level, which persists regardless of beat detection algorithm improvements. It captures the case where spectral SNR can appear acceptable while individual beat timing is too imprecise for HRV.

When palm routing is triggered, the reason is logged (composite score failure vs. amplitude floor trigger) so the clinical interface can distinguish between general signal quality problems and the specific physics limitation of face measurement on darker skin.

4. Pilot Evaluation

4.1 Subjects and Protocol

Two subjects participated in pilot evaluation sessions. Subject 1 (ITA range 4–10, FST V) and Subject 2 (ITA range –35 to –5, FST V–VI). Sessions were conducted in a standard home office environment under ceiling lighting without controlled illumination. No special equipment was used beyond a standard integrated laptop webcam (640×480, 30 fps nominal).

Each session consisted of a 10-second calibration phase followed by a 35-second recording. Subjects were instructed to remain as still as possible but were not constrained. Ground truth HR was measured simultaneously with a fingertip pulse oximeter (SpO2 device) at 1-second intervals.

A total of 12 sessions were recorded across both subjects, varying lighting conditions, time of day, and deliberate motion conditions (one session per subject with intentional coughing and repositioning during recording).

4.2 Heart Rate Accuracy

HR agreement between PulseRoute and the pulse oximeter reference is summarized in Table 1. All sessions where the FFT cross-check did not fire (8 of 12) produced HR estimates within ± 3 BPM of the reference. Sessions where the cross-check fired (4 of 12) showed the sub-harmonic selection problem before correction; after correction, agreement was within ± 4 BPM.

Table 1. HR accuracy across pilot sessions

Condition	Sessions	Mean Error (BPM)	Max Error (BPM)
Still, good lighting	4	1.4	2.5
Still, dim lighting	4	2.1	3.8
Motion (cough/reposition)	4	3.2	6.1
All sessions	12	2.2	6.1

4.3 Respiratory Rate

Respiratory rate was successfully detected in 10 of 12 sessions with confidence above 0.7. The two failures occurred during sessions with deliberately slow breathing (6–7 BrPM), which falls at the edge of the detection range. In all successful detections, the adaptive notch filter correctly removed the breathing artifact from the pulse signal, with HR band power ratio improving from a mean of 10.3% pre-filter to 86.7% post-filter.

4.4 Routing Behavior

The routing system was evaluated across sessions designed to test both routing paths:

Face accepted: 7 of 12 sessions. All had filtered std > 0.002 and composite score above the ITA-adjusted threshold. Mean HR error in these sessions: 1.8 BPM.

Palm routed — amplitude floor: 3 of 12 sessions. All were FST V–VI subjects under dim or standard home lighting, with filtered std 0.0007–0.0016. The composite score in two of these sessions actually passed the threshold (score 0.42–0.49, threshold 0.50 for FST V), demonstrating that the amplitude floor check catches cases the composite score misses.

Palm routed — composite score: 2 of 12 sessions. Both were high-motion sessions where RR regularity and spectral SNR both degraded.

False palm routings (face signal was actually valid): 0 of 12 sessions. The system did not incorrectly route any session where face measurement was reliable.

4.5 HRV and RMSSD

RMSSD was not validated against ECG in this pilot (pulse oximeter provides HR but not beat-to-beat intervals). Qualitative observations: RMSSD values in face-accepted sessions with filtered std > 0.003 ranged 60–150 ms, consistent with literature norms for resting healthy adults. Sessions below the amplitude floor showed RMSSD of 200–360 ms — confirming the inflation pattern that motivated the floor threshold. All such sessions were correctly routed to palm by the amplitude floor check.

4.6 Motion Robustness

In the two deliberate-motion sessions, the motion rejection system rejected 9.6% and 37% of frames respectively (the latter being sustained deliberate movement). In both cases the system successfully accumulated 35 seconds of clean signal within the 87.5-second wall-clock cap. The recording extended to 47 seconds and 84 seconds respectively. HR error in these sessions was 3.2 and 6.1 BPM — the higher error in the sustained motion session attributable to the signal degradation from motion artifacts that survived frame-level rejection and entered the FFT.

5. Discussion

5.1 The Inclusivity Gap in rPPG

Our results confirm the documented performance gap for darker skin tones in rPPG systems. The fundamental cause is physical: melanin absorption reduces reflected light amplitude before it reaches the camera sensor, and no signal processing algorithm can recover information that was never captured. The correct clinical response to this physical limitation is not to try harder with the face signal — it is to route to a measurement site that does not have this limitation. The palm, as shown by Luo et al. [8], provides a consistently stronger signal regardless of skin tone because palmar skin contains minimal melanin even in FST VI individuals.

PulseRoute's contribution is not that it solves the physics problem, but that it correctly identifies when the physics problem is present and responds appropriately. The amplitude floor at 0.002 filtered std is a principled threshold derived from empirical data, not an arbitrary cutoff. It corresponds to the point at which individual beat timing errors exceed the physiologically meaningful variation in RR intervals, making RMSSD measurement meaningless.

5.2 Per-Session Calibration vs. Population Thresholds

The SubjectProfile architecture has a fundamental advantage over population-average thresholds that goes beyond skin tone. Any factor that changes the signal amplitude — lighting quality, camera quality, distance from camera, ambient temperature (which affects skin perfusion) — is automatically accommodated because all thresholds are derived from the patient's own signal in their own environment. A patient with a poor webcam in dim lighting gets a motion threshold calibrated to their own noise floor, not to a research-grade camera under 500 lux.

This design philosophy — anchor all thresholds to the individual, not the population — is directly applicable to any biometric system deployed in uncontrolled environments, and represents a generalizable contribution beyond the rPPG domain.

5.3 Limitations

Small pilot sample: Two subjects with 12 sessions provides initial validation but is insufficient for clinical claims. A larger study with ECG ground truth, diverse skin tones, ages, and medical conditions is required before deployment.

No palm validation: The palm pipeline exists in the codebase (Step 2 palm ROI, Step 3 palm signal extraction) but was not evaluated in this pilot. The routing system correctly identifies when to route to palm but the palm signal quality has not been measured against ground truth.

RMSSD not ECG-validated: Pulse oximeter ground truth provides HR but not beat-to-beat intervals. RMSSD accuracy requires ECG or a validated PPG device with beat-detection capability.

Single camera type: All sessions used a standard 640×480 laptop webcam. Performance on higher-resolution cameras or IR-capable cameras has not been characterized.

Age range not covered: Both subjects were adults aged 20–30. Age affects both HRV norms and skin optical properties; older subjects have not been tested.

5.4 Clinical Deployment Considerations

PulseRoute is designed for asynchronous teleconsultation where the measurement runs during the first 45–50 seconds of a session. The 10-second calibration phase is presented to the patient as "the system is preparing to measure" — no user action is required. The subsequent 35-second recording presents a visible countdown with motion quality indicators. If palm routing is triggered, the interface asks the patient to show their palm to the camera.

One important design decision is that palm routing is never forced — the patient can decline and the system will report face measurement results with an explicit low-confidence flag. This preserves patient autonomy while communicating uncertainty to the clinician.

6. Conclusion

We presented PulseRoute, a contactless vital signs monitoring system designed for inclusive teleconsultation. The system's two novel contributions — per-session subject calibration and adaptive face-to-palm routing — address the principal failure modes of existing rPPG systems when deployed on diverse populations in uncontrolled environments. Pilot evaluation on two FST V–VI subjects demonstrates correct HR estimation, accurate respiratory rate detection, and appropriate palm routing triggered by the physics-based amplitude floor in conditions where existing face-only systems would produce silently inflated HRV measurements without clinical warning.

The routing architecture — face first for user experience, palm as fallback for signal quality — is, to our knowledge, the first published implementation of a systematic face-to-palm rPPG routing system with a principled signal quality decision boundary. The system is open source and the complete pipeline is publicly available at <https://github.com/arijbelmabrouk/PulseRoute>.

Future work will focus on: (1) large-scale validation with ECG ground truth across all six Fitzpatrick types; (2) palm signal quality characterization and end-to-end routing validation; (3) integration with real teleconsultation platforms; and (4) extension to SpO₂ estimation using dual-wavelength approaches that do not require infrared hardware.

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